EE-354 SYNCRONOUS GENERATOR

E/21/291

1. What is an electrical Machine?

Electrical machines can be defined as devices that converts electrical energy in to mechanical energy or vice versa. most commonly motors and generators are examples of electrical machines. Moreover electrical machines are of 2 types AC and DC machines.

2. What are the 2 main components of a synconous generator?

Stator - stationary part consist of armature winding
Rotor - Rotating Part generates the magnetic field.

3. What is the main difference between syncronous machine and induction machine?

Syncronous machines operate at syncronous speed, which means that the rotor speed is equal to the speed of rotating magnetic field.

Induction motors and generators operate on electromagnetic induction. The rotor does not get power directly. It rotates due to the induced current caused by the stators rotating magnetic field.

4. What are the type of rotors used in synconous generators?

Cylindrical rotor - Cylindrical steel shaft with embeded windings. used for high speed generators and motors.

Salient pole rotor - poles of the generator are projected outwards. used in low speed generators such as in hydro power stations.

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What are the methods used to supply DC voltage to the rotor winding?

Slip rings and brushes - DC current flows to the rotor field winding through carbon brushes that makes contact with slip rings.

Brushless and exitation - A small exitation generator is mounted on the same shaft as the main generator.

6. what is the drawback of slip rings and brushes method?

The carbon brushes wear down over time and needed to be replaced. Sparking can occur at high speeds and mechanical contact creates heat.

7. How is voltge generated in stator windings when the load is zero?

When the rotor is exited with DC current and rotated at synchronous speed the magnetic flux created by it cuts the stator windings. Therefore according to faraday's law The changing magnetic field induces a voltage in the stator.

8. what are the armature reactions?

Armature reaction is the effect of magnetic field produce by the stator on the main magnetic field produced by the ~~stator~~ rotor. This tends to magnetize and demagnetize the main field.

9. what are the types of losses in a synchronous generator?

copper losses - heat generated in the stator and rotor
core losses - hystresis and eddy current
mechanical losses - friction in bearings
Stray losses -

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what are the tests carried out to find parameters of a synchronous generator?

Open circuit test
Short circuit test
DC test

11. what data are available on the nameplate of a synchronous generator?

Rated voltge
Rated current
Rated power
Power factor
frequancy
speed
manufacturer details

12. what are the advantages of operating synchronous generators in parallel?

improved reliability
load sharing
Ease of maintanance
Improved operating efficiency.

13. what methods are used to identify whether phase sequence is the same?

Phase sequance indicator
Three lamp method
Two lamp method

14. what is a synchronoscope?

A synchronoscope is an instrument used to indicate whether the incoming generator is synchronized with the bus bar by checking phase sequence phase angle difference.

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